

Network Parameters: S-parameters

Theory, Definition, applications, and Analysis

Contents

I	Ports and terminals	2
1	Definition of ports	3
1.1	Prologue	3
1.2	Concept of black box	3
1.3	Concept of Ports vs Terminals	3
1.4	Formalizing notation	4
II	Scattering parameters and port equations	5
2	Multi-port network parameters	6
2.1	Prologue	6
2.2	s-parameter	6
2.2.1	Example: a 2-port frequency domain s-parameter	7
2.3	Scattering parameter and port waves	7
2.3.1	Voltage waves and voltage scattering parameters	8
2.3.2	Current waves and current scattering parameters	9
2.3.3	The wave amplitude s-parameter	9
2.3.4	The power waves s-parameter	9
III	References	11

Part I

Ports and terminals

1. Definition of ports

1.1 Prologue

A network can be viewed as a design comprising distinct circuitry units or specific geometric structures, each engineered to fulfill a particular functionality within a larger overall system or design.

Often, our primary focus is on understanding and analyzing the interaction between two networks. Depending on the application, the internal structural details of these networks—such as internal components and circuitry—may not be relevant. This information might be unknown, concealed, or intentionally disregarded. In some cases, the designs do not contain actual internal circuitry; instead, they rely solely on specific geometries or physical shapes to achieve the desired electromagnetic functionality. Examples of such networks include high-speed interconnects, antennas, waveguides, mixers, and duplexers, which are commonly used in microwave technology.

1.2 Concept of black box

Here, we formalize key definitions that readers may encounter in the literature. These concepts have been primarily introduced within the field of system identification.

Black box: A network N is referred to as a "black box" when its internal structure and components are inaccessible or unknown [1].

White box: The white-box concept is the exact opposite of the black box approach, referring to systems or designs where every internal detail is known.

From a circuit design perspective, a white-box design is one for which the complete "netlist" is available.

Gray box: A gray-box system represents networks or systems where only limited information about their internals is available or accessible.

1.3 Concept of Ports vs Terminals

As mentioned above, in many cases, our primary focus is solely on studying and analyzing the bilateral interaction between two networks connected through specific terminals

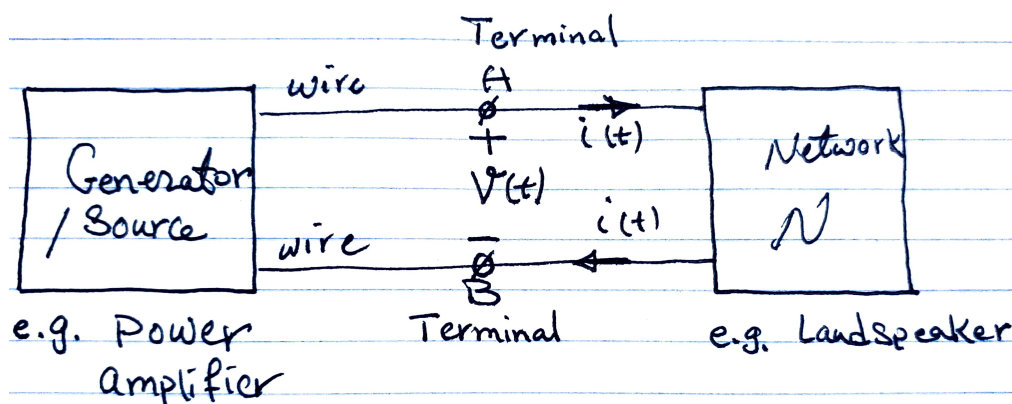


Figure 1: A generator (source) delivering power to a network N through its terminals.

In electrical network theory, according to the following definition, a two-terminal network N , as shown in Figure 1, is referred to as a "one-port" network if

Port: A pair of terminals in a circuit where, at all times, the instantaneous current flowing into one terminal is equal to the instantaneous current flowing out of the other terminal [1].

1.4 Formalizing notation

The concept of a port has proven invaluable in the analysis of hierarchical designs that incorporate black box modules. Here, we formalize notation to ensure uniformity throughout the manuscript.

- $i(t)$: the current entering the port is called *port current*
- $v(t)$: the voltage across the port is called *port voltage*
- $p(t)$: the *instantaneous power* delivered to the network through the port is $p(t) = v(t)i(t)$. It is defined according to the reference directions in Figure 1
- $W(t)$: The energy delivered to the one-port network from time t_0 to t is

$$W(t) \triangleq \int_{t_0}^t p(t') dt' = \int_{t_0}^t v(t') i(t') dt' \quad (1)$$

For voltage and current measured in volts (V) and amperes (A), respectively, power is measured in watts (W) and energy in joules (J).

Part II

Scattering parameters and port equations

2. Multi-port network parameters

2.1 Prologue

For illustration purpose, consider a two-port device/subcircuit $\mathcal{N}$ connected between four nodes in a circuit as

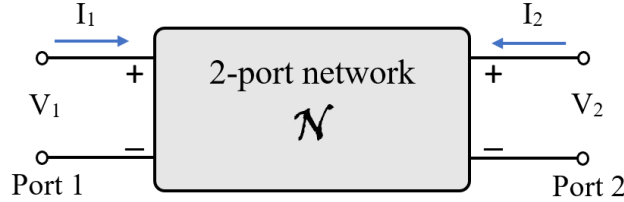


Figure 2: General two-port network.

The port-to-port behavior (response) of a two-port network in Figure 2 is defined in terms of the four port variables V_1 , I_1 , V_2 , and I_2 that are the voltages and currents at the terminals of port-1 and port-2 respectively. The H-, Y-, and Z- network parameters are based on relating the total voltages and total currents at the ports. In the microwave frequency range, reflection and transmission behavior of networks is characterized using s-parameter *viz.* by relating the traveling (incident and reflected) waves at networks' ports (as shown in Figure 3 and Figure 5).

2.2 s-parameter

The transition from voltages and currents to wave quantities requires considering a network accessible solely through ports and hence can be only portwise connected [5] to other blocks.

We consider an arbitrary port# i (Figure 3) with its instantaneous voltage V_i and current I_i .

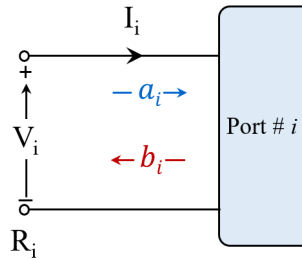


Figure 3: (Arbitrary) i^{th} port of a network with voltage V_i , current I_i , incident wave a_i , reflected wave b_i .

To define the instantaneous wave quantities a_i and b_i , we assign a constant positive port resistance R_i to this port. The incident and reflected waves a_i and b_i , respectively, are defined (/quantified) by

$$\begin{aligned} a_i &= \frac{1}{2} (V_i + R_i I_i) \\ b_i &= \frac{1}{2} (V_i - R_i I_i) \end{aligned} \quad (2)$$

Strictly speaking, the quantities defined by (2), having the dimension of voltage (volts), are so-called voltage waves¹.

The s-parameter representation of a subnetwork is given based on the traveling waves as

$$s_{ii} = \frac{b_i}{a_i} \quad (3)$$

2.2.1 Example: a 2-port frequency domain s-parameter

The S-parameters of a two port network as illustrated in Figure 4 in frequency domain (FD) is given by (4).

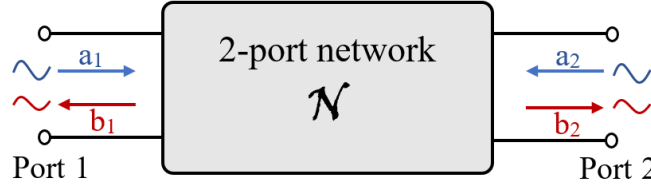


Figure 4: Incident voltage waves (a_1, a_2) and reflected voltage waves (b_1, b_2) used in s-parameter definitions of two-port network.

$$\begin{bmatrix} b_1(s) \\ b_2(s) \end{bmatrix} = \begin{bmatrix} S_{11}(s) & S_{12}(s) \\ S_{21}(s) & S_{22}(s) \end{bmatrix} \begin{bmatrix} a_1(s) \\ a_2(s) \end{bmatrix} \quad (4)$$

where $S_{ij}(s)$ relates input j to the output i at every frequency points. In turn, it is defined as the ration between the reflected voltage (/power) wave traveling outward from i^{th} port and the incident voltage (/power) wave entering to i^{th} port. To this end, the reflected wave at port j is measured when it is induced only by the incident wave at port i where all the other ports are terminated with their characteristic impedance.

2.3 Scattering parameter and port waves

Figure 5 illustrates the port# i of a network characterized by its s-parameter. To avoid confusion one can consider it as a single port ($i = 1$) or may assume that its all the ports are terminated only with their characteristic impertinences. Therefore, the incident (forward-traveling) a_i and reflection (backward-traveling) b_i waves at the port are induced only by the source connected to this node E_i and i.e. $s_{ii} = \frac{b_i}{a_i}$.

In Figure 5, V_i^f and I_i^f are incident voltage and current waves, respectively and V_i^b and I_i^b are reflection voltage and current waves, respectively.

¹Later, in we will elaborate on this concept by introducing current- and voltage- wave amplitude, and the power waves transmitted through ports.

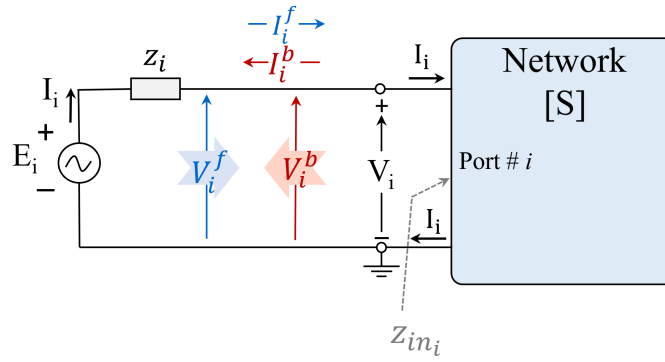


Figure 5: i^{th} port of a network with termination, incident (/forward), and reflection (/backward) waves.

Practical considerations² allow the following assumptions:

1. The resistive reference impedance z_i is a resistor with positive value, i.e., $z_i = R_i$
2. The port input impedance $Z_{in_i} = R_i$

The following relationships can be investigated in Figure 5.

$$\begin{cases} I_i = I_i^f - I_i^b, \\ V_i = V_i^f - V_i^b, \end{cases} \quad \Rightarrow \quad \begin{cases} I_i^f = I_i + I_i^b, \\ V_i^f = V_i + V_i^b, \end{cases} \quad \text{and} \quad \begin{cases} I_i^b = I_i^f - I_i, \\ V_i^b = V_i^f - V_i \end{cases} \quad (5)$$

For the matching condition with real terminations as $Z_i = Z_{in_i} = R_i$, it is $V_i^f = \frac{E_i}{2}$

2.3.1 Voltage waves and voltage scattering parameters

As primarily presented in Section 2.2, the quantities in (6) are so-called voltage waves and have the dimension of voltage.

$$\begin{aligned} V_i^f &= \frac{E_i}{2} = \frac{1}{2} (V_i + R_i I_i) \\ V_i^b &= V_i - \frac{V_i + R_i I_i}{2} = \frac{1}{2} (V_i - R_i I_i) \end{aligned} \quad (6)$$

Hence, the s-parameter defined based on the voltage waves in (6) is

$$S_{11}^V = \frac{V_i^b}{V_i^f} = \frac{V_i - R_i I_i}{V_i + R_i I_i} \quad (7)$$

²Considering that the typical characteristic impedance of general transmission lines in microwave frequency range 50Ω , in practice, for the s-parameter measurement/computation the typical values for this reference (/normalization) impedance is chosen to be $z_i = R_o = 50\Omega$.

At microwave frequencies, most transmission lines have a 50-ohm characteristic impedance. It is to be noted that 50Ω is a typical value and lines are usually built to have a well defined characteristic impedance of 50Ω . However, lines of 75-, 90-, and 300-ohm impedance are also designed for specific cases.

2.3.2 Current waves and current scattering parameters

In a similar way as presented in Section 2.3.1, we can also define the current waves that have the dimension of current (amperes) as shown in (8) in below.

$$\begin{aligned} I_i^f &= \frac{E_i}{2R_i} = \frac{V_i + R_i I_i}{2R_i} \\ I_i^b &= \frac{V_i + R_i I_i}{2R_i} - I_i = \frac{V_i - R_i I_i}{2R_i} \end{aligned} \quad (8)$$

Hence, the s-param defined based on the current waves in (8) is

$$S_{11}^I = \frac{I_i^b}{I_i^f} = \frac{V_i - R_i I_i}{V_i + R_i I_i} \quad (9)$$

Observation: There is not much practical value in adopting the current wave quantities. By comparing (7) and (9), it is apparent that

$$S_{11}^I = S_{11}^V \quad (10)$$

2.3.3 The wave amplitude s-parameter

By comparing (9) and (9) in sections 2.3.1 and 2.3.2, respectively, It is

$$S_{11} = \frac{b_i}{a_i} = \frac{V_i - R_i I_i}{V_i + R_i I_i} \quad (11)$$

Hence, specifying "current wave" or "voltage wave" is not needed. The s-parameter definition as given in (11) is generally referred to as **the wave amplitude s-parameter**.

2.3.4 The power waves s-parameter

The power-waves are defined according to the direction of the power flow as [4, Sec. 1.2] and [2, 3]

$$\begin{cases} a_i = \frac{V_i + z_i I_i}{2\sqrt{|\operatorname{Re} z_i|}} & \left(\frac{V}{\sqrt{\Omega}} \right) \end{cases} \quad (12a)$$

$$\begin{cases} b_i = \frac{V_i - z_i^* I_i}{2\sqrt{|\operatorname{Re} z_i|}} & \left(\frac{V}{\sqrt{\Omega}} \right) \end{cases} \quad (12b)$$

where

- a_i and b_i are normalized complex voltage waves incident on and reflected from i^{th} port,
- V_i is the voltage at i^{th} port and I_i is the current flowing into that port,
- z_i is any arbitrary reference impedance at the i^{th} port.

By choosing the (real-valued) resistance as the reference (/normalization) impedance $z_i = R_i = 50 \Omega$ simplifies the form of (12) to

$$\begin{cases} a_{pi} = \frac{V_i + R_i I_i}{2\sqrt{R_i}} \\ b_{pi} = \frac{V_i - R_i I_i}{2\sqrt{R_i}} \end{cases} \quad (13a)$$

$$\begin{cases} b_{pi} = \frac{V_i - R_i I_i}{2\sqrt{R_i}} \end{cases} \quad (13b)$$

For a general multiport network illustrated in Figure 6, forming the s-parameters matrices is straightforward by using a_i and b_i from (12).

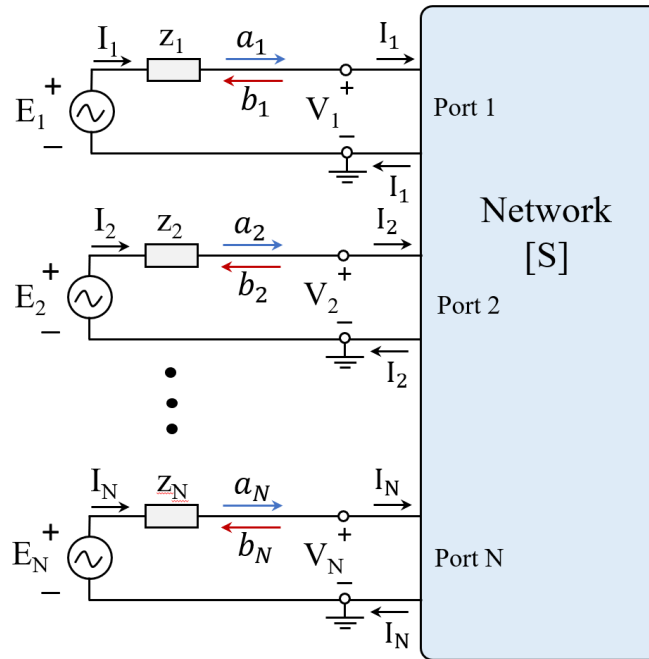


Figure 6: Multiport network whose port-to-port behavior was characterized with s-parameter matrix transfer function.

Part III

References

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